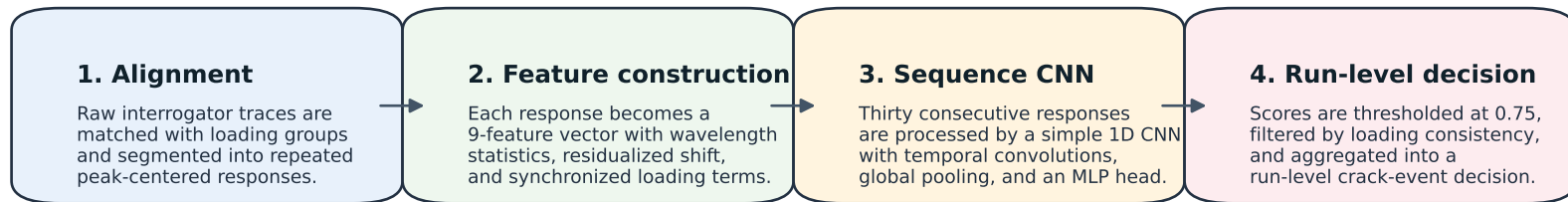


Method Overview

Peak-aligned FBG responses are converted into compact loading-aware sequences and classified under strict run-level validation.



Retained operating point

Leave-one-run-out validation; sequence length = 30; prediction horizon = 5; seed = 42; weighted BCE training; physics-consistency filter with mean force slope and maximum force thresholds.

Final 9-feature set

wl2_median, wl2_std, delta_wl_ch2, force_N, displacement_mm, air_pressure_bar, delta_wl_rate, delta_disp_rate, is_small_sample